
VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION–LANGUAGE MODELS

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ABSTRACT

Vision–Language Models (VLMs) provide a broadly useful basis for zero-shot goal-conditioned value functions, yet frozen pre-trained representations can restrict transfer and temporal reasoning when progress must be inferred from trajectories. We introduce VITA, a zero-shot value function learning method that expands the usable scope of such representations through test-time adaptation. At inference, a lightweight adaptation module is updated with a gradient step on a meta-learned self-supervised loss, so that each update is trained to improve value estimation rather than merely fit an auxiliary task. Sequential updates over a trajectory store history in the module parameters, allowing temporal context to be incorporated without changing the underlying VLM encoder. To reduce reliance on dataset-specific shortcuts, we use a dissimilarity-based sampling strategy that selects semantically diverse trajectory segments during training. Across real-world robotic manipulation evaluations, VITA transfers from a single training environment to out-of-distribution tasks, scenes, and embodiments, outperforming the state-of-the-art zero-shot approach based on autoregressive VLMs. We further show that these zero-shot value estimates can serve as reward shaping signals for offline reinforcement learning, yielding multi-task policies on Meta-World that outperform policies trained with the benchmark’s fuzzy-logic dense rewards.

1 INTRODUCTION

Vision-Language Models (VLMs) have shown that large-scale, weakly structured web data can support broad generalization across tasks and domains without task-specific human supervision (Radford et al., 2021; Alayrac et al., 2022). By contrast, generalist policies for robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024) still depend heavily on expert demonstrations (Ho & Ermon, 2016), which limits how easily they can scale across behaviors and settings. World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) offer one route toward broader agent learning by generating simulated experience for domains such as robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). As these models become capable of producing realistic visual trajectories in increasingly diverse and open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), the central question becomes more general: *how can agents extract scalable learning signals from videos without relying on human supervision?*

One route learns policies directly from expert visual trajectories by inferring latent actions from visual transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a). Such latent actions can in principle be mapped to executable controls, though this mapping remains difficult in practical deployments. A complementary route seeks a broadly applicable *goal-conditioned value function* that can be reused as a zero-shot reward-shaping signal for reinforcement learning (RL) or as supervision for imitation learning (Chen et al., 2021; Ma et al., 2022). Under this view, goal-conditioned value estimation asks a general progress-estimation question: given visual observations and a natural-language task description, how far has the agent advanced toward completing the specified task? (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025).

Prior work has used pre-trained contrastive VLMs (Radford et al., 2021) for zero-shot reward shaping by interpreting similarity between task descriptions and visual observations in a shared multimodal space as semantic progress (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024). This

framing is appealing because it can be applied across tasks, but it lacks the temporal context needed to distinguish states that look similar while representing different stages of an activity, such as folding versus unfolding a shirt. Autoregressive VLMs (Alayrac et al., 2022) offer a different route by conditioning on an entire visual trajectory in the prompt; however, when used for progress estimation, they can inherit a bias toward monotonically increasing predictions from chronologically ordered pre-training data (Ma et al., 2024). Thus, both contrastive and autoregressive VLM approaches depend on fixed pre-trained representations for zero-shot prediction (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023), which can constrain transfer and temporal reasoning in trajectory-based value estimation (Pătrăucean et al., 2023). Recent efforts broaden value-function learning through large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), and demonstration-based reward specification (Sontakke et al., 2023).

In this work, we introduce VITA, a zero-shot value function learning method that uses test-time adaptation to extend frozen VLM representations to trajectory-level progress estimation, outperforming the state-of-the-art zero-shot approach (Ma et al., 2024) on our real-world robotic manipulation evaluations. Rather than requiring additional large-scale pre-training or task-specific demonstrations at deployment, VITA adapts online to the semantic and temporal context of each trajectory. At inference, a lightweight adaptation module is updated in negligible time with a gradient step on a meta-learned self-supervised loss (Sun et al., 2024), such that each update is trained to improve value estimation (Finn et al., 2017). Sequential updates over a test trajectory encode history into the estimator parameters (Sun et al., 2024), providing temporal context while preserving chronological order. To reduce shortcut learning (Geirhos et al., 2020), we introduce dissimilarity-based sampling, which encourages training on semantically varied trajectory segments. Our evaluation covers three uses of a broadly applicable value estimator within the evidence supported by the paper: transfer under task, environment, and embodiment shifts; discrimination between expert and non-expert trajectories; and reward shaping for offline RL (Levine et al., 2020).

We summarize our key contributions as follows:

- We propose VITA, a test-time adaptation method that broadens contrastive VLMs into zero-shot value estimators with stronger transfer and temporal reasoning, without requiring task-specific demonstrations or additional large-scale pre-training.
- VITA transfers from a single training environment to varied out-of-distribution tasks, environments, and robot embodiments in robotic manipulation, outperforming the state-of-the-art zero-shot method (Ma et al., 2024).
- VITA’s zero-shot value estimates also function as broadly usable reward-shaping signals, producing offline RL policies on the Meta-World multi-task benchmark (MT10) that surpass policies trained with the simulation’s fuzzy-logic dense rewards (Yu et al., 2020).

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

For visual trajectories, task progress estimation can be cast as goal-conditioned value function estimation whenever task completion defines the relevant reward. We therefore learn a goal-conditioned value function by predicting completion progress from vision–language representations. Formally, the value function is defined as: $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$ which maps an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar value indicating predicted progress toward the goal. We set $V(o_t; g) = 0$ at task start and $V(o_t; g) = 1$ at completion. In expert demonstrations, progress is commonly approximated by temporal position under the assumption that successful trajectories move monotonically toward the goal (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, temporal progress is defined using normalized timestep indices: $V^{\pi_E}(o_t; g) = \frac{t}{T}$ where T is the trajectory length. This temporal proxy supplies supervision for learning the goal-conditioned value function V without requiring manually specified dense rewards.

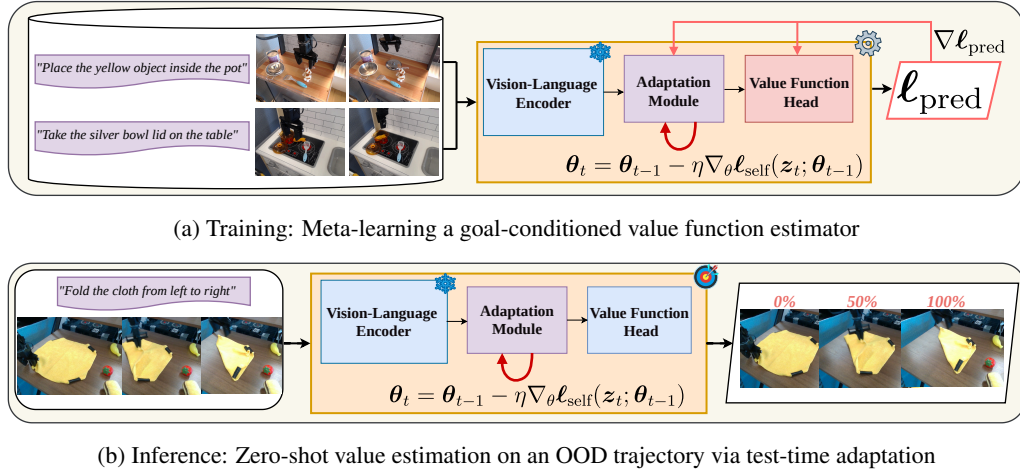


Figure 1: **Overview of VITA.** VITA meta-learns a goal-conditioned value function and uses test-time adaptation to transfer zero-shot to out-of-distribution trajectories.

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) treats inference itself as a self-supervised learning problem (Goodfellow et al., 2016), updating model parameters on each test instance without labels. Although it was first introduced for static image classification (Sun et al., 2020), recent extensions apply the same principle to sequence modeling (Sun et al., 2024; Wang et al., 2023), where a parametric adaptation module f_{adapt} receives a gradient update at every timestep. Unlike standard sequence models that store history in activations, such as recurrent hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key-value pairs (Vaswani et al., 2017), TTT writes trajectory information into the parameters of f_{adapt} through sequential inference-time updates. These parameters act as an implicit memory that preserves temporal order while accumulating context. Sun et al. (2024) further show that the self-supervised task can be meta-learned under the gradient-based meta-learning paradigm (Finn et al., 2017). In that setting (Sun et al., 2024), learnable linear projections parameterize the self-supervised loss, which is optimized for downstream prediction after an update rather than for auxiliary reconstruction alone. We adapt this broader mechanism to goal-conditioned value learning, using sequential updates to capture temporal context and meta-learning the self-supervised task so that adaptation improves value estimation.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

The goal-conditioned value function estimator is organized into three reusable components: (1) a multimodal encoder that extracts joint vision-language representations from trajectories and task descriptions; (2) an adaptation module that is updated at test time with a self-supervised loss meta-learned for value estimation; and (3) a regression head that maps the adapted representation to value estimates.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We instantiate the multimodal encoder with a frozen contrastive vision-language model, CLIP (Radford et al., 2021), which provides representations for both visual observations and goal descriptions. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we concatenate the corresponding embeddings at each timestep t to obtain joint multimodal representations $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, and ϕ_v and ϕ_g denote the visual and language encoders, respectively. Because CLIP is pre-trained to align paired image-text inputs in a shared representation space (Radford et al., 2021), observations that are semantically closer to goal completion are expected

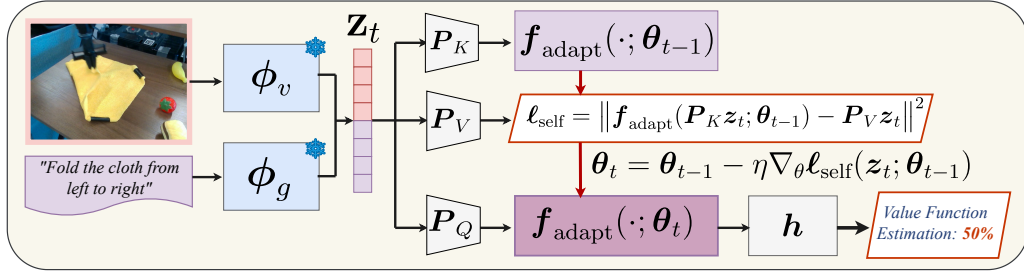


Figure 2: **Test-time Adaptation.** During inference, each timestep t updates the adaptation module f_{adapt} with a gradient step on a meta-learned self-supervised loss ℓ_{self} , allowing temporal history to be encoded in the module parameters.

to lie closer to the goal description representation. This property makes the encoder a general semantic substrate, while the adaptation module supplies trajectory-specific context.

3.1.2 TEST-TIME ADAPTATION

To make multimodal representations responsive to both semantic content and trajectory context, we use an adaptation module f_{adapt} following the test-time training paradigm described in 2.2. At inference, f_{adapt} is updated online by a gradient step on the meta-learned self-supervised loss ℓ_{self} . The loss is a reconstruction objective built from learnable linear projections P_K and P_V , which are trained so that adaptation improves value estimation on downstream progress labels. In particular, given $P_K \in \mathbb{R}^{d' \times 2d}$ that generates a perturbed input view and $P_V \in \mathbb{R}^{d' \times 2d}$ for the target, the self-supervised loss is defined as:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} before test-time adaptation. At each timestep t , the parameters are updated by

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . Because these updates are applied in chronological order, f_{adapt} retains information from previous observations in its parameters and thereby supplies temporal context to the value estimator. Section 4.6.2 shows empirically that this incremental update pattern is more effective than performing adaptation only at the trajectory level.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After test-time adaptation, a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps the current input z_t into the adaptation space $\mathbb{R}^{d'}$. The adapted representation then passes through f_{adapt} and a regression head h , producing the goal-conditioned value estimate:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), following value-function estimator architectures used in deep reinforcement learning (Espeholt et al., 2018). Training uses normalized progress labels y_t from expert demonstrations as described in Section 2.1, with a supervised prediction loss ℓ_{pred} defined as the mean squared error between predicted values $V(z_t; g)$ and targets y_t . At inference, h remains frozen while f_{adapt} updates online using the self-supervised objective. This separation allows the estimator to reuse the same learned value head while adapting its internal representation to new trajectories without task-specific demonstrations.

3.2 TRAINING PROCEDURE

We train the value function estimator with gradient-based meta-learning, optimizing the self-supervised loss ℓ_{self} so that test-time adaptation reduces the supervised prediction loss ℓ_{pred} (Sun et al.,

2024). Rather than expose the model only to adjacent, highly redundant frames, training uses diverse sub-trajectories selected by dissimilarity-based sampling. This sampling choice encourages value estimates to rely on semantic and temporal evidence that is more likely to transfer across trajectories.

3.2.1 META-LEARNING

During training, the adaptation module f_{adapt} is updated online at each timestep using the self-supervised loss ℓ_{self} . Following gradient-based meta-learning (Finn et al., 2017), we differentiate through these updates with respect to the supervised prediction loss ℓ_{pred} . This jointly optimizes the initialization θ_0 of f_{adapt} , the linear projections P_K, P_V, P_Q , and the regression head h . Formally, the total training loss combines ℓ_{pred} and ℓ_{self} , weighted by a scalar λ . The learned θ_0 becomes the starting point for inference-time adaptation, so the model is trained not only to predict progress on the training trajectories but also to adapt productively when presented with new ones.

3.2.2 DISSIMILARITY-BASED SAMPLING

Expert visual trajectories commonly contain many adjacent frames with little visual change, a redundancy also observed in video action recognition (Wang et al., 2018). If training batches overemphasize such repeated frames, the value estimator may exploit shortcuts (Geirhos et al., 2020; Lee et al., 2021b), for example by associating frequently occurring late-stage appearances with high progress. To make the training signal less tied to dataset-specific frame redundancy, we construct mini-batches from visually diverse sub-trajectories within each trajectory. The resulting dissimilarity-based sampling increases intra-batch variation and acts as an importance-sampling heuristic that favors semantic diversity.

Formally, given a sequence of multimodal representations $\{z_1, \dots, z_T\}$, we extract fixed-length sub-trajectories using a sliding window of size w_{tr} and stride s , yielding a candidate set $\mathcal{W}$. A diverse subset of size k maximizes pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i is a sub-trajectory of length w_{tr} . Solving Eq. 4 is intractable, requiring enumeration of all $\binom{|\mathcal{W}|}{k}$ subsets. To approximate this objective efficiently, we adopt a scoring-based heuristic, assigning each window $w \in \mathcal{W}$ a score equal to its total dissimilarity with all others:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

By computing a diversity score s for each window and selecting the k highest-scoring windows, we obtain a heuristically diverse subset with polynomial-time complexity. The additional cost is small relative to model training, as analyzed in Appendix G. The ablation in Section 4.6.1 compares this sampling strategy with full-trajectory sampling and shows that the diversity-biased batches improve discrimination between expert and non-expert visual trajectories.

4 EXPERIMENTS

Our evaluation tests whether VITA supports capabilities expected of a broadly useful goal-conditioned value function: (i) value estimation for real-world robotic manipulation under shifts in task, environment, and embodiment; (ii) discrimination between expert and non-expert visual trajectories in the same robotic domain; and (iii) zero-shot reward shaping for offline RL on Meta-World MT10, a simulated benchmark for multi-task robot learning.

4.1 TRAINING SETUP

We train VITA on expert visual trajectories paired with natural-language task descriptions from BridgeData V2 (Walke et al., 2023), a dataset that contains manipulation tasks collected across multiple environments and robot embodiments. To make transfer measurable, our training subset is deliberately narrower than the full benchmark: it contains 2,986 expert demonstrations of pick-and-place tasks and excludes folding, sweeping, and stacking. All training demonstrations use a single

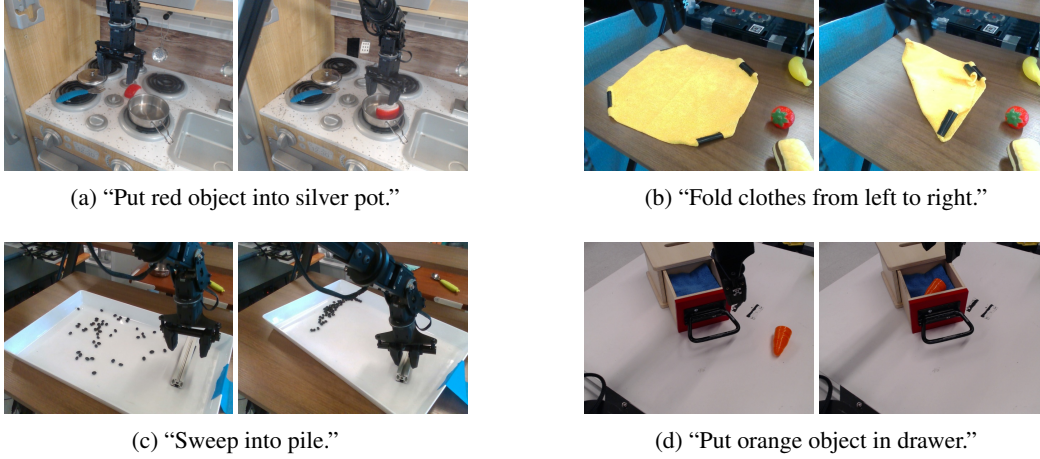


Figure 3: Examples of visual trajectories paired with task descriptions across the distribution shifts considered in our evaluation. (a) In-distribution. (b, c) Environment shift. (d) Combined embodiment and environment shift.

WidowX 250 embodiment across 4 ToyKitchen configurations. We reserve 287 additional expert demonstrations as the in-distribution test set, denoted `tk_pnp`; Appendix A shows examples from the training set. We use OpenCLIP ViT-B/32 as the frozen backbone for encoding video frames and task descriptions. The training objective combines ℓ_{pred} with ℓ_{self} , weighted by $\lambda_{\text{self}} = 0.5$. During training, dissimilarity-based sampling uses window size $w_{\text{tr}} = 8$, number of sub-trajectories $k = 8$, and stride $s = 1$. At test time, we apply one gradient step ($t_{\text{ep}} = 1$) using λ_{self} with learning rate $\eta = 0.1$. Appendix D reports implementation details and hyperparameter sweeps.

4.2 BASELINES

We compare VITA with value estimators that instantiate different ways of using contrastive and autoregressive VLMs. VLM-CL produces zero-shot values from cosine similarity between CLIP representations of each frame and the task description (Baumli et al., 2023). VLM-RM regularizes CLIP embeddings by projecting features along the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT trains a supervised regression head on frozen CLIP multimodal features, while CLIP-GRU adds a Gated Recurrent Unit (GRU) (Cho et al., 2014) over the same frozen features. GVL (Ma et al., 2024) is the state-of-the-art zero-shot value function based on autoregressive VLMs. In our experiments, GVL-0S denotes the zero-shot setting, where the prompt contains only the test trajectory and task description. GVL-1S denotes a one-shot in-context setting, where a full shuffled trajectory from the training distribution and its progress labels are provided as an example. Following GVL, we use Gemini 1.5 Pro (`gemini-1.5 Pro-latest`) (Gemini Team et al., 2024) with its proposed prompt template. We also tested Qwen-VL 2.5 (Yang et al., 2024), which did not overcome temporal bias, and GPT-4o (Hurst et al., 2024), which refused to produce value estimates. Appendix D provides additional implementation details.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

We evaluate whether progress estimators transfer beyond the training conditions by using BridgeData V2 subsets curated to vary task, environment, and robot embodiment. Environment shifts change scene layout or background: for example, `lm_pnp` is a pick-and-place task in front of a laundry machine, while `td_fold`, `ft_fold`, and `rd_fold` place cloth folding on different surfaces. `ms_sweep` further tests a long-horizon sweeping task in a confined tray. Embodiment shifts use the DeepThought robot, which differs from the training WidowX 250 embodiment in both morphology and camera perspective. `dt.tk_pnp` (pick-and-place) and `dt.tk_stack` (stacking) keep the in-distribution environment but change embodiment, whereas `dt.ft_stack` (stacking) and `dt.rd_pnp` (drawer pick-and-place) combine embodiment and environment shifts. Each dataset contains 200 expert trajectories with task descriptions, except `ms_sweep`, which contains 100 because of dataset size.

Table 1: VOC scores for value estimation across task, environment, and embodiment shifts. **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Both Shifts.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
ES	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	<u>0.658</u>
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

Appendix B gives the full dataset description. We evaluate value estimates with *Value Order Correlation* (VOC) (Ma et al., 2024), which measures how well predicted progress aligns with frame chronology in a trajectory. Formally, let $p_1, \dots, p_T$ denote predicted progress values for a trajectory of T frames, and let $r_t = t$ represent the temporal index. VOC is the Spearman rank correlation ρ_s between predicted values and temporal indices.

Table 1 shows that VITA and CLIP-GRU consistently outperform GVL-0S and GVL-1S, indicating that temporal modeling over frozen multimodal features transfers more reliably in this evaluation than prompting pre-trained autoregressive VLMs for progress estimates. Among the temporal models, VITA outperforms CLIP-GRU on 6 of 10 evaluation datasets. Both methods also outperform CLIP-based baselines without temporal modeling (CLIP-FT, VLM-CL, VLM-RM), emphasizing that progress estimation requires more than static semantic similarity. CLIP-FT is competitive with GVL on the in-distribution task but does not maintain comparable performance under out-of-distribution conditions. For GVL-0S and GVL-1S, in-context examples sometimes help, sometimes have little effect, and sometimes degrade performance. Both GVL variants perform relatively well on folding tasks (td_fold, ft_fold, rd_fold) but not as well on stacking (dt_tk_stack, dt_ft_stack) and pick-and-place (dt_tk_pnp, lm_pnp), suggesting that the autoregressive VLM used in GVL may transfer unevenly across manipulation types. By contrast, VITA maintains high performance across the evaluated task categories and distribution shifts, supporting the broader applicability of test-time adaptation for zero-shot value estimation within this benchmark suite.

The long-horizon ms_sweep task remains challenging for most methods, but VITA obtains the highest VOC score, including an improvement over CLIP-GRU. This result suggests that parameter-based temporal memory can be especially useful when progress unfolds over longer visual sequences. Under embodiment shifts, VITA also retains strong performance. On dt_tk_pnp, where the task and environment match training but the robot changes, both VITA and CLIP-GRU exceed their in-distribution scores, indicating that learned temporal context can transfer across embodiments in this setting. In contrast, GVL-1S performs poorly on both dt_tk_pnp and dt_tk_stack, showing that the evaluated one-shot prompting setup does not reliably address embodiment shifts.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond transfer across datasets, we test whether VITA assigns progress estimates that distinguish expert behavior from suboptimal behavior in the same visual domain. We compare predictions on expert and scripted non-expert trajectories collected from the in-distribution BridgeData V2 setting. The scripted trajectories are generated in ToyKitchen with a heavily randomized controller (Walke et al., 2023) under the same embodiment and environment configuration as the expert demonstrations. They include pick-and-place tasks over object categories that may overlap with training, including general objects (sc_pnp_obj), rigid items (sc_pnp_robj), soft toys (sc_pnp_stoy), utensils (sc_pnp_uten), and a cloth (sc_pnp_cloth). Thus, the comparison isolates behavioral quality more than broad visual domain shift: the scripted trajectories match the training setting in task family, embodiment, and environment, but deviate from the smooth monotonic progress typical of

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-0S	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: (a) Discrimination between expert and scripted robot demonstrations, measured by average BinVOC across 5 in-distribution scripted datasets. (b) Offline RL performance on Meta-World MT10, measured by interquartile mean (IQM) with 95% stratified bootstrap CIs.

expert demonstrations. Appendix C shows scripted examples. Because the randomized controller is not calibrated to produce graded levels of suboptimality, we do not interpret these data as a continuous quality scale. Instead, a useful value estimator should assign lower VOC to scripted trajectories than to expert trajectories from the same setting. We measure this with *BinVOC*, defined as $\mathbf{1}[\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, which equals 1 when expert trajectories receive higher VOC than suboptimal ones and 0 otherwise.

We report discriminative performance as average BINVOC over the scripted datasets. Expert performance (VOC_{exp}) is measured on the in-distribution test set `tk_pnp`, whose pick-and-place tasks are comparable to those in the scripted evaluation. VITA, GVL-0S, and GVL-1S achieve perfect discrimination by assigning higher VOC to expert trajectories for every scripted task. VITA also outperforms CLIP-GRU, suggesting that recurrent hidden-state memory may overfit temporal shortcuts more readily than implicit memory formed by sequential test-time updates. The CLIP-based zero-shot baselines (VLM-CL, VLM-RM) likely underperform because they lack temporal modeling, while CLIP-FT is more reliable but fails on one task, indicating sensitivity to object-level distribution changes.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

We evaluate whether the learned value estimator can be reused as a reward signal beyond the real-world BridgeData setting by applying it to Meta-World MT10 (Yu et al., 2020). MT10 contains ten robotic manipulation tasks, including pick-and-place, door opening, and pushing. In this experiment, VITA is trained on real-world robotic data and evaluated zero-shot as a dense reward model in simulation. For each Meta-World MT10 task, we generate 20 expert visual demonstrations with the benchmark’s expert policies. The goal-conditioned value estimator defines a dense reward for each visual observation, and policies are trained with Implicit Q-Learning (IQL) (Kostrikov et al., 2021) in the offline RL setting. We repeat expert data generation and policy training across 10 random seeds. After training, each multi-task policy is evaluated on 20 rollouts per task, and we compute the average return for every (task, seed) pair. Following Agarwal et al. (2021), we report the interquartile mean (IQM) over all pairs with 95% stratified bootstrap confidence intervals. Appendix E describes the training and evaluation protocol in more detail.

Table 2 reports MT10 performance for the baselines under this protocol. VITA achieves the highest IQM return (0.815), outperforming all CLIP-based baselines. We do not compare directly with GVL at this scale because it relies on Gemini-1.5, a proprietary autoregressive VLM whose inference cost is prohibitive for RL reward labeling, and the open-source alternatives tested in preliminary experiments did not produce reliable progress estimates. We also include Meta-World’s fuzzy-logic dense reward (META-WL), which achieves 0.779. VITA outperforming this task-engineered reward suggests that a value estimator trained on real-world visual demonstrations can provide useful reward shaping in a simulated multi-task benchmark.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We test whether dissimilarity-based sampling broadens the training signal by comparing sub-trajectory sampling strategies on the expert-versus-non-expert discrimination task. Full-trajectory sampling (Sun et al., 2024) computes the adaptation loss over all overlapping sub-trajectories with stride $s = 1$, which in our setup encourages reliance on global temporal shortcuts. Random sampling increases coverage by selecting sub-trajectories uniformly, but it does not explicitly target semantic variation. Dissimilarity-based sampling instead favors sub-trajectories with high pairwise dissimilarity, yielding stronger discrimination between expert and non-expert visual trajectories. Appendix F.1 provides the full comparison.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We compare adaptation strategies that incorporate temporal context at different scopes. VITA uses step-by-step adaptation without reset, so parameters are incrementally updated from θ_{t-1} and serve as implicit memory. This design preserves temporal order while accumulating trajectory history. We compare it with three alternatives: (i) *trajectory-level adaptation (TTT-TR)*, which applies a single gradient update over the full trajectory and therefore averages losses across frames without preserving order; (ii) *memoryless adaptation (TTT-RS)*, which resets the adaptation module to θ_0 at every step and adapts only to the current visual observation; and (iii) *explicit memory (TTT-EX)*, which also resets the adaptation module at every step but updates it with a local window of recent frames matching the meta-training window size ($w_{tr} = 8$). Across the evaluated datasets, VITA consistently outperforms trajectory-level, memoryless, and explicit-memory variants, showing that incremental parameter memory is the most effective temporal mechanism in our setup. Appendix F.2 gives the full analysis.

5 RELATED WORK

5.1 VLMs AS VALUE FUNCTION ESTIMATORS

Contrastive VLMs have been used as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024) by scoring frame-level similarity, but this static formulation does not model temporal context. Other work fine-tunes CLIP on domain-specific video datasets (Fan et al., 2022; Jiang et al., 2024) to obtain dense rewards, which can improve a target domain but reduces zero-shot applicability. RoboCLIP (Sontakke et al., 2023) supports one-shot reward specification without fine-tuning, yet it provides sparse rewards and still requires demonstrations. Large-scale multimodal pre-training methods (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) learn language-conditioned value functions by relying on scale for transfer, while ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data with video rewinding and a small number of in-domain demonstrations. VITA instead adapts a pre-trained contrastive vision–language encoder at inference, using test-time adaptation to add temporal and semantic context without additional large-scale pre-training or domain-specific demonstrations. More generally, the method is designed for pre-trained vision–language representations that can support universal value estimation, with meta-learned adaptation broadening their temporal reasoning and transfer. Autoregressive VLMs (Alayrac et al., 2022) provide another path by leveraging in-context learning over trajectories and have been applied to success detection (Du et al., 2023) and goal-conditioned value functions (Ma et al., 2024). GVL (Ma et al., 2024) addresses monotonic bias from ordered pre-training data by shuffling frames at inference, but this removes chronological order, which is important for distinguishing visually similar states at different task stages. VITA preserves the observed order while incorporating trajectory context through test-time adaptation.

5.2 TEST-TIME ADAPTATION

Test-time adaptation offers a parameter-efficient way to specialize VLMs to a current input or context without full fine-tuning. Prompt-based methods optimize model parameters at test time using task context, although the mechanisms behind the learned prompts can be difficult to interpret (Zhou et al.,

2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) argue that CLIP representations can support such adaptation, motivated by evidence that CLIP encodes human-interpretable concepts identified through mechanistic interpretability (Bricken et al., 2023). CLIP-based similarity has also been used as a reward signal while optimizing parameters at test time with reinforcement learning (Zhao et al., 2024). Test-time training (Sun et al., 2020) provides a related but more sequence-oriented mechanism: an adaptation module is updated with a self-supervised loss at each timestep. Unlike meta-reinforcement learning methods (Finn et al., 2017; Bauer et al., 2023), which require task-specific adaptation episodes during training, test-time training performs online adaptation during inference without task labels, making it a natural fit for zero-shot progress estimation from trajectories.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

Our results show that test-time adaptation can turn frozen vision–language representations into value functions that transfer across task, environment, and embodiment shifts in robotic manipulation. The same estimator also distinguishes expert from non-expert trajectories and supplies reward shaping for offline RL, suggesting that the learned progress signal is useful beyond a single evaluation protocol. By updating sequentially over a trajectory, VITA stores history in its parameters and captures temporal and semantic context more effectively than static CLIP-based baselines and autoregressive VLM prompting in our experiments. Its gains over CLIP-GRU on expert–non-expert discrimination and offline RL further suggest that implicit memory through sequential test-time updates can transfer better than explicit recurrent hidden states for zero-shot value estimation. Pre-trained autoregressive VLMs such as GVL retain broad semantic knowledge, but they are not explicitly trained for progress estimation, which requires temporally grounded reasoning over visual trajectories. In our setup, per-timestep adaptation adds negligible inference overhead because only a lightweight module is updated, preserving practical applicability for the evaluated scenarios.

6.2 LIMITATIONS AND FUTURE WORK

Although test-time adaptation improves zero-shot progress estimation across the unseen tasks and environments evaluated here, the method may still be challenged by trajectories with high execution variability, long durations, or safety-critical deployment constraints. The adaptation module is lightweight, but updating a value estimator at every timestep may be undesirable in some real-time closed-loop systems, especially when guarantees about online parameter changes are required. Future work should therefore study alternative adaptation schedules and safety-aware variants, including uses of vision–language adaptation for agents trained inside world models. Applying VITA to real-time closed-loop control and more complex RL environments is another natural extension. Because the method is agnostic to the particular vision–language encoder, it may also improve temporal reasoning for other domain-specific multimodal representations that already encode useful value information. Finally, our experiments show that dissimilarity-based sampling reduces shortcut learning empirically, but a theoretical account of why diversity-biased sampling changes shortcut reliance remains open. This analysis is outside the scope of our core contribution: a value estimator that meta-learns a test-time adaptation mechanism to improve zero-shot performance and temporal reasoning.

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A TRAINING DATASET

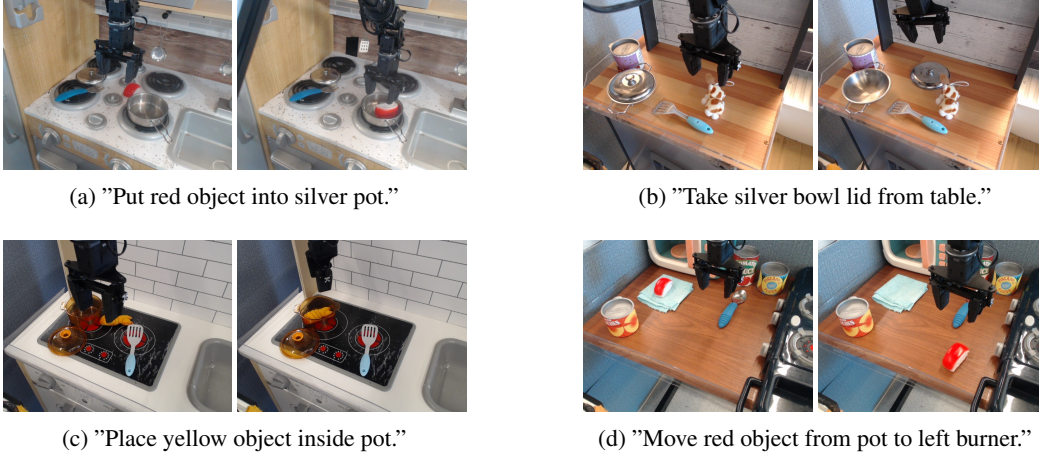


Figure 4: Each subfigure shows start and end frames from a training expert demonstration paired with its natural-language task description. The demonstrations span four ToyKitchen configurations while remaining within the pick-and-place training scope.

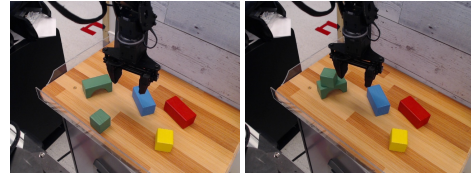
B EVALUATION DATASETS

Table 3: Evaluation datasets organized by task type, environment, and robot embodiment.

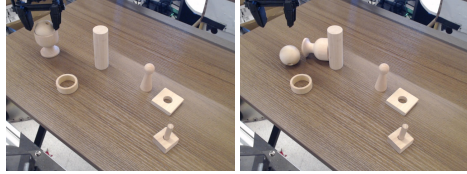
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

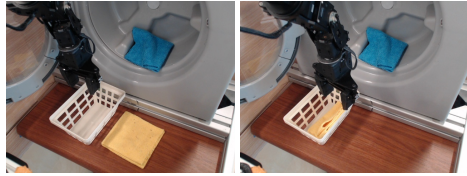


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

Figure 5: Each subfigure shows start and end frames from an evaluation trajectory that changes robot embodiment relative to training, paired with its natural-language task description. The top row keeps the ToyKitchen environment while using the DeepThought robot, whereas the bottom row combines the embodiment shift with new environments.



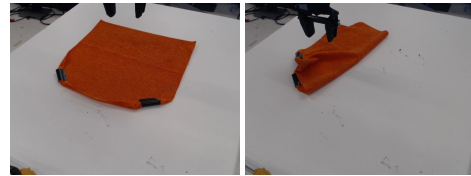
(a) lm_pnp: "Place blue cloth inside washer."



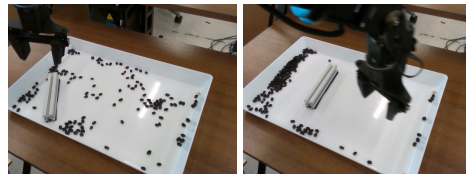
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Each subfigure shows start and end frames from an evaluation trajectory that changes environment relative to training, paired with its natural-language task description. The robot embodiment is fixed, so these examples isolate visually and structurally different scenes.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Each subfigure shows start and end frames from a scripted evaluation trajectory collected with the WidowX 250 robot in ToyKitchen. The tasks remain within pick-and-place behavior but vary the object category, and we use generic task descriptions because public annotations are unavailable.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

We use the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset as a frozen backbone, so all adaptation occurs outside the base vision-language encoder. Each visual observation and task description is encoded with the CLIP vision and text encoders, respectively. Based on validation performance, we use a joint concatenation-based representation rather than an element-wise product. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activation and projection dimension $d' = 64$. The model is trained with AdamW using learning rate 1×10^{-4} , weight decay 1×10^{-4} , and a cosine learning-rate schedule with 10% warmup. We use batch size 32 and pad trajectories to 120 frames, matching the longest training sequence. The self-supervised loss weight λ_{self} is set to 0.5 based on validation performance. We train for 5 epochs because validation VOC typically plateaus early and extended training gives no further improvement. Unless otherwise specified, dissimilarity-based sampling uses stride $s = 1$, sub-trajectory length $w_{\text{tr}} = 8$, and $k = 8$ selected windows. All experiments were run on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference, only the temporal adaptation module f_{adapt} is updated, using the same projection dimension ($d' = 64$) as in training. We use one gradient step ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$. This setting is selected from a sweep over learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation steps $\{1, 5, 10\}$, with validation results shown in Table 4. Both a single larger update ($\eta = 0.1$, $t_{\text{ep}} = 1$) and multiple smaller updates ($\eta = 0.01$, $t_{\text{ep}} = 10$) achieve comparable validation

Table 4: Validation VOC for VITA under different test-time learning rates η and adaptation-step counts t_{ep} .

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

performance. We choose the single-step setting because it is slightly stronger and keeps inference adaptation minimal.

D.3 BASELINES

VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rate $\eta = 0.1$, selected by the validation sweep described above. CLIP-FT keeps the same overall value-prediction architecture but removes the adaptation module and self-supervised loss. It uses a frozen CLIP encoder, followed by a linear projection and two-layer MLP trained with standard supervised regression, without meta-learning or test-time adaptation. To give this non-adaptive baseline sufficient capacity, it uses an $8\times$ larger projection matrix and $10\times$ more training steps than our method (Lin et al., 2022). For VLM-RM, we use a baseline prompt describing the environment. CLIP-GRU also keeps the frozen CLIP encoder and regression head h , but replaces f_{adapt} with a single-layer GRU of hidden size 64 whose final hidden state is passed to h . For both GVL-0S and GVL-1S, we use Gemini 1.5 Pro, gemini-1.5-pro-latest (Gemini Team et al., 2024), whereas the original GVL implementation used gemini-1.5-pro. We also evaluated GVL with the open-source autoregressive VLM Qwen-VL 2.5 (Yang et al., 2024), which did not overcome temporal bias despite frame shuffling, while GPT-4o (Hurst et al., 2024) consistently declined to produce scalar progress estimates in our setup. All evaluations use the prompt template introduced in the original GVL work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

We use Implicit Q-Learning (IQL) with one configuration shared across all MT10 tasks. Expectile regression uses $\tau = 0.7$, advantage weighting temperature is 3.0, and the clipping threshold is 100.0. The actor and critic learning rates are 1×10^{-4} and 3×10^{-4} , respectively, with batch size 256. Policies train for 100,000 gradient steps and are evaluated every 10,000 steps on 20 rollouts per task with horizon 150. The implementation follows d3rlpy, with configuration: expectile=0.7, weight_temp=3.0, max_weight=100.0, actor_lr=1e-4, critic_lr=3e-4, batch_size=256.

E.2 EVALUATION PROTOCOL

Following Agarwal et al. (Agarwal et al., 2021), we report the interquartile mean (IQM) of returns across all task–seed pairs. IQM averages the middle 50% of scores, reducing sensitivity to outliers relative to the mean or median. We estimate 95% confidence intervals with 10,000 stratified bootstrap resamples, stratified by task. The same evaluation procedure is applied to every method in the Meta-World MT10 experiments.

Table 5: Disc@VOC across scripted datasets for training-time sampling strategies used in test-time adaptation. A checkmark indicates successful expert-versus-suboptimal discrimination. Sampling methods **FT** (full-trajectory), **Rand** (random sampling), and **Diss** (dissimilarity-based sampling) are parameterized by (w_{tr}, k) .

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We evaluate how broadly the sampling strategy supports discriminative adaptation by comparing several ways to choose sub-trajectories during training. Table 5 reports Disc@VOC on 5 scripted datasets, indicating whether a model assigns higher average VOC to expert trajectories (`tk_pnp`) than to each scripted dataset. We compare three strategies: (1) full-trajectory sampling, which applies adaptation over all overlapping sub-trajectories of length w_{tr} with stride s , following prior work (Sun et al., 2024); (2) random sampling, which uniformly selects k sub-trajectories of length w_{tr} per video; and (3) dissimilarity-based sampling, which constructs candidate sub-trajectories of length w_{tr} with stride s and selects k windows that maximize pairwise dissimilarity in representation space.

Dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ achieves perfect discriminative performance. Selecting fewer windows ($k = 4$) of the same length also succeeds, whereas shortening the windows to $w_{tr} = 4$ reduces effectiveness. Full-trajectory sampling performs worst, consistent with the concern that adapting over all consecutive windows can encourage global temporal shortcuts. Random sampling improves over full-trajectory sampling but is less reliable than dissimilarity-based selection. These results support the broader claim that both semantic diversity and local trajectory structure matter for discriminative test-time adaptation.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

Table 6: Validation VOC for progress estimation under alternative temporal-adaptation strategies and distribution shifts. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

We compare four adaptation scopes: (1) TTT-TR, which performs one offline gradient update over the full trajectory. This batched update averages losses across frames and does not preserve temporal

order; (2) TTT-RS, which resets the adaptation module at every step and updates only on the current visual observation, preventing temporal information from carrying forward; (3) TTT-EX, which also resets at every step but updates with a local window of recent frames matching the meta-training window size ($w_{\text{tr}} = 8$). This averages gradients over the window and discards within-window order; (4) VITA (OURS), which updates incrementally at every timestep and forms implicit memory by accumulating temporal history in the adapter parameters. We use the same test-time hyperparameter sweeps for all variants, as described in Appendix D. VITA, TTT-RS, and TTT-TR perform best with learning rate 0.1 and one adaptation step, whereas TTT-EX performs best with learning rate 1.0 and one adaptation step.

Table 6 shows that TTT-TR, TTT-RS, and TTT-EX obtain similar performance. Batched updates over either a short explicit window (TTT-EX) or the full trajectory (TTT-TR) therefore behave similarly to the memoryless baseline (TTT-RS) in this evaluation. VITA consistently outperforms all three, indicating that the relevant scope for temporal adaptation is the ordered, per-timestep update sequence. With implicit memory, updating over a local frame window would incorporate the most recent observations twice: explicitly through the window and implicitly through the accumulated parameters. Updating less frequently would also omit intermediate frames. Thus, in this setting, implicit-memory test-time training benefits from sequential per-timestep updates without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ denote the candidate set with size $N_c = |\mathcal{W}|$. To approximate diverse subset selection without enumerating all $\binom{N_c}{b}$ subsets as in Eq. 4, we use a scoring heuristic that assigns each candidate window a diversity score equal to the sum of its pairwise distances to all other windows, i.e., the row sum of the distance matrix. With trajectory length T , the number of candidate windows is $N_c = T - w_{\text{tr}} + 1$. Each window is represented as a flattened feature vector of dimension $m = w_{\text{tr}} \cdot d$, where d is the dimension of each multimodal representation z_t . We compute the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU using batched `cdist`, which requires $\mathcal{O}(N_c^2 \cdot w_{\text{tr}} \cdot d)$ operations.

In our experiments, training trajectories are short, with mean length 69 and maximum length 120. For batch size 32, feature dimension $d = 1024$ for the CLIP-based multimodal representation, and window length $w_{\text{tr}} = 8$, the worst-case computational cost is:

$$32 \cdot N_c^2 \cdot w_{\text{tr}} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This overhead is negligible relative to a single model forward and backward pass, while enabling selection of diverse windows that broaden the training signal.